

Flight Readiness Review — Payload Thermal Analysis (v2)

High-Altitude Balloon Event-Camera Payload
Operating concept: on-board computer powered at 16 km

Thermal Analysis — Systems Engineering

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Abstract

This report documents the thermal model and flight-readiness assessment of the high-altitude balloon (HAB) event-camera payload for the operating concept in which the on-board computer (OBC) is powered only on reaching the imaging altitude of 16 km. A coupled three-block model (external environment, enclosure-wall conduction, and a two-dimensional planar internal field) is solved over the ascent-to-apogee mission window from a cold-night initial temperature of $+5\text{ }^{\circ}\text{C}$. The internal cavity is modelled as a planar (x, z) cross-section of the full 170 mm interior width, capturing the left-right asymmetric component layout without invoking axial symmetry. The component set reflects the mechanical CAD bill of materials; storage peripherals (SSD, USB stick, SD card reader) absent from the CAD are excluded. All ten payload components remain within their operating temperature limits throughout the analysed window. The minimum cold-side margin is $+15.0\text{ }^{\circ}\text{C}$ and the tightest hot-side margin is $+8.7\text{ }^{\circ}\text{C}$ (Wolfgang GA100). The thermal design is assessed **GO** for flight.

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1 System Description and Geometry

The payload is housed in a cylindrical expanded-polypropylene (EPP) Thermobox enclosure. The internal cavity is modelled as a two-dimensional planar (x, z) section of full width W_{int} and height H_{int} , extruded to an out-of-plane slab depth D_{int} for the volumetric energy balance. The outer enclosure shell is physically cylindrical, and the outer geometry $(R_{box}, D_{box}, L_{box})$ is retained for

the external convection calculation in Block A. The internal field is solved on a Cartesian (x, z) mesh covering the full cavity width from left wall to right wall; no axial-symmetry assumption is applied, and both lateral walls appear as independent Dirichlet boundaries. The geometry is summarised in Table 1.

Table 1: Enclosure geometry. Internal dimensions describe the planar meshed domain; external dimensions are used for the Block A convection calculation only.

Symbol	Quantity	Value
W_{int}	Internal cavity width (planar x -extent)	0.170 m
H_{int}	Internal cavity height	0.250 m
D_{int}	Out-of-plane slab depth	0.170 m
L_{wall}	Side-wall thickness (EPP)	0.030 m
L_{cap}	End-cap thickness (EPP)	0.030 m
R_{box}	Outer radius (convection reference)	0.115 m
L_{box}	Outer height	0.310 m
D_{box}	Outer diameter (convection)	0.230 m

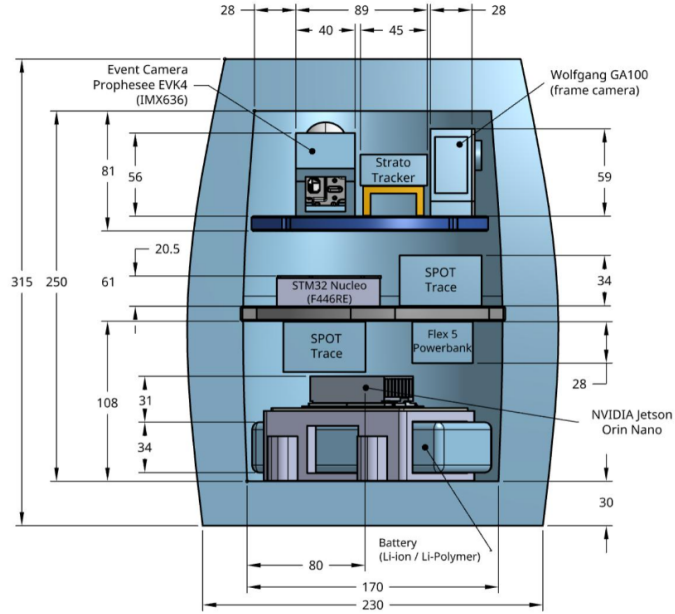


Figure 1: Mechanical CAD section used as the reference layout for the meshed model. The interior resolves four stacked bays: the camera bay on top (EVK4 left, Strato Tracker centre, GA100 right), two mid-deck rows (STM32 upper-left with SPOT Trace, Flex 5 powerbank lower-right with SPOT Trace), and the bottom bay (battery below in a PETG cage, Jetson carrier board and SoC above in a stacked topology). Components are mounted with clearance to the side walls.

2 Modelling Methodology

The model is decomposed into three coupled blocks. Block A provides the external free-stream temperature and convective coefficient as a function of altitude and ascent velocity. Block B solves transient conduction through the enclosure wall and end caps. Block C solves the two-dimensional planar temperature field of the internal cavity, driven by the component heat loads

and bounded by the Block B wall temperatures. The blocks are advanced with a lagged coupling each timestep.

2.1 Block A — External Environment and Forced Convection

The ambient state is evaluated with the US Standard Atmosphere 1976, piecewise by layer. Within a layer of base altitude h_b , base temperature T_b , base pressure P_b and lapse rate L_b ,

$$T(h) = T_b + L_b (h - h_b), \quad (1)$$

$$P(h) = \begin{cases} P_b \left(\frac{T}{T_b} \right)^{-g/(L_b R)}, & L_b \neq 0, \\ P_b \exp\left(-\frac{g(h - h_b)}{R T_b}\right), & L_b = 0, \end{cases} \quad \rho = \frac{P}{R T}, \quad (2)$$

with $g = 9.80665 \text{ m/s}^2$ and $R = 287.058 \text{ J/(kg K)}$. Transport properties use Sutherland-type fits,

$$\mu(T) = \frac{1.458 \times 10^{-6} T^{3/2}}{T + 110.4}, \quad k_{air}(T) = \frac{2.495 \times 10^{-3} T^{0.8646}}{1 + 245.4 \cdot 10^{-12}/T/T}, \quad \text{Pr} = \frac{\mu c_p}{k_{air}}. \quad (3)$$

The enclosure is treated as a cylinder in cross-flow at the ascent velocity V . The Reynolds number and the Churchill–Bernstein Nusselt-number correlation give the external convective coefficient:

$$\text{Re} = \frac{\rho V D_{box}}{\mu}, \quad (4)$$

$$\text{Nu} = 0.3 + \frac{0.62 \text{Re}^{1/2} \text{Pr}^{1/3}}{[1 + (0.4/\text{Pr})^{2/3}]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282\,000} \right)^{5/8} \right]^{4/5}, \quad h_{ext} = \frac{\text{Nu} k_{air}}{D_{box}}. \quad (5)$$

2.2 Block B — One-Dimensional Wall Conduction

Each wall segment (side, top cap, bottom cap) is discretised into N nodes across the thickness, with spacing $\Delta x = L_{wall}/(N - 1)$. Transient conduction is advanced with implicit Euler in a finite-volume form. Boundary nodes carry half control volumes; interior nodes carry full control volumes. The outer surface (node 0) exchanges with the atmosphere through h_{ext} and the inner surface (node $N - 1$) with the cavity air through the internal coefficient h_{int} :

$$\text{Outer: } \frac{\rho c_p \Delta x}{\Delta t} (T_0^{n+1} - T_0^n) = h_{ext} (T_\infty - T_0^{n+1}) + \varepsilon \sigma (T_{sink}^4 - (T_0^{n+1})^4) + \frac{k}{\Delta x} (T_1^{n+1} - T_0^{n+1}), \quad (6)$$

$$\text{Interior: } \frac{\rho c_p \Delta x}{\Delta t} (T_i^{n+1} - T_i^n) = \frac{k}{\Delta x} (T_{i-1}^{n+1} - 2T_i^{n+1} + T_{i+1}^{n+1}), \quad (7)$$

$$\text{Inner: } \frac{\rho c_p \Delta x}{\Delta t} (T_{N-1}^{n+1} - T_{N-1}^n) = \frac{k}{\Delta x} (T_{N-2}^{n+1} - T_{N-1}^{n+1}) + h_{int} (T_{air} - T_{N-1}^{n+1}). \quad (8)$$

The outer surface exchanges long-wave radiation with the environment in addition to convection. The net radiative flux $\varepsilon \sigma (T_{sink}^4 - T_0^4)$ is linearised about the previous outer-surface temperature as $h_{rad} (T_{sink} - T_0^{n+1})$ with $h_{rad} = 4 \varepsilon \sigma (T_0^n)^3$, preserving the tridiagonal structure. The mission flies at night, so no absorbed solar flux is applied. The outer-surface emissivity is $\varepsilon = 0.90$ (EPP shell) and the effective radiative-sink temperature is $T_{sink} = 216.65 \text{ K}$ (-56.5°C), taken at the stratospheric ambient minimum of the US Standard Atmosphere 1976. These equations form a tridiagonal system solved directly each timestep. The camera optical windows are modelled as a thin glass pane solved with the same one-dimensional scheme and coupled to the internal field as a parallel conductance on the upper boundary.

The internal coefficient h_{int} represents natural convection of the cavity air. In the laminar enclosure regime the Nusselt number scales as $Nu \propto Ra^{1/4}$, and since $Ra \propto \rho^2 \propto P^2$ the coefficient varies with the square root of the local ambient pressure $P(h)$ from Block A:

$$h_{int}(h) = h_{int,0} \sqrt{\frac{P(h)}{P_0}}, \quad h_{int,0} = 1.5 \text{ W}/(\text{m}^2 \text{ K}), \quad P_0 = 101,325 \text{ Pa}. \quad (9)$$

As the payload ascends and the atmosphere thins, the internal convective coupling weakens, decreasing from $1.5 \text{ W}/(\text{m}^2 \text{ K})$ at sea level to approximately $0.14 \text{ W}/(\text{m}^2 \text{ K})$ near apogee.

2.3 Block C — Two-Dimensional Planar Internal Field

The cavity interior is modelled as a planar (x, z) cross-section of width W_{int} and height H_{int} , extruded to out-of-plane depth D_{int} . The mesh is uniform with $N_x \times N_z$ rectangular cells; cell (i, j) has its centre at $x_i = (i + \frac{1}{2})\Delta x$ and $z_j = (j + \frac{1}{2})\Delta z$, with $\Delta x = W_{int}/N_x$ and $\Delta z = H_{int}/N_z$.

The face areas and cell volume are uniform across all cells:

$$A_x = \Delta z D_{int}, \quad A_z = \Delta x D_{int}, \quad V_P = \Delta x \Delta z D_{int}. \quad (10)$$

There is no radial weighting; the formulation is purely Cartesian. Each cell P is advanced with an implicit finite-volume energy balance over its four faces f to neighbours F , with internal source Q_P :

$$\frac{\rho_P c_{p,P} V_P}{\Delta t} (T_P^{n+1} - T_P^n) = \sum_f G_f (T_F^{n+1} - T_P^{n+1}) + Q_P. \quad (11)$$

Interface conductances use the harmonic-mean conductivity to handle dissimilar materials,

$$G_f = \frac{k_{harm} A_f}{d}, \quad k_{harm} = \frac{2 k_P k_F}{k_P + k_F}, \quad (12)$$

where $d = \Delta x$ for x -faces (left/right) and $d = \Delta z$ for z -faces (top/bottom).

All four domain boundaries are Dirichlet conditions driven by Block B inner-surface temperatures. The left wall ($i = 0, x = 0$) and right wall ($i = N_x, x = W_{int}$) each receive the inner side-wall temperature profile $T_w(z)$; the bottom cap and top cap receive T_{bot} and T_{top} respectively. A boundary face uses the half-cell distance $d/2$, giving

$$G_{bc} = \frac{k A_f}{d/2}, \quad (13)$$

contributing G_{bc} to the diagonal and $G_{bc} T_{bc}$ to the right-hand side. On the upper boundary the glass window adds a parallel conductance path of area A_{win} to T_{window} alongside the EPP wall path. Because neither lateral wall is a symmetry axis, the full left-right asymmetric component layout is captured directly without mirroring or halving the domain. Assembling all $N_x N_z$ cells yields a sparse linear system $\mathbf{A} \mathbf{T}^{n+1} = \mathbf{b}$, solved with a sparse direct solver each timestep.

2.4 Coupling

Per timestep n : (i) Block A returns T_∞ , h_{ext} and the ambient pressure from the altitude and velocity profiles; (ii) the cavity air temperature T_{air} is taken as the volume-weighted mean of the air cells and h_{int} is evaluated at the current altitude; (iii) Block B advances the side wall, caps and window pane; (iv) Block C advances the internal field using the updated inner-surface temperatures. The numerical parameters are listed in Table 2.

Table 2: Numerical and boundary parameters.

Symbol	Quantity	Value
$N_x \times N_z$	Internal mesh (planar cells)	68×100 (6 800 cells)
$\Delta x, \Delta z$	Cell size	2.5 mm
N	Wall nodes (Block B)	5
Δt	Timestep (implicit)	30 s
t_{end}	Analysis window	6800 s (113.3 min)
$h_{int,0}$	Internal coefficient (sea level)	1.5 W/(m ² K)
ε	Outer-surface emissivity (radiation)	0.90
T_{sink}	Radiative-sink temperature	216.65 K (−56.5 °C)
T_{init}	Initial temperature	278.15 K (+5 °C)

3 Component Database

The payload comprises ten active thermal regions (Table 3). The component set is taken directly from the mechanical CAD bill of materials; storage peripherals (250 GB SSD, USB 3.0 stick, SD card reader) that do not appear in the CAD section are excluded from this analysis. The Jetson Orin Nano is represented by two mesh regions: a passive carrier board and an active SoC/heatsink sub-region that carries the full 7 W electrical dissipation, consistent with the right-biased SoC placement visible in the CAD section (Fig. 1). OBC-gated loads (*Gated: Yes*) are unpowered during ascent and energised on reaching 16 km. Table 4 lists the operating limits and the data source for each power and limit value.

Table 3: Component database: material, mass, steady dissipation, and OBC-gating flag. Gated components draw power only during the 16 km OBC window.

Component	Subsys.	Material	m [g]	P [W]	Gated
Battery (Li-ion / Li-Po)	EPS	Li-ion cell	1000	0.30	—
Jetson Orin Nano (carrier board)	OBC	FR4/ICs	180	0.00	—
Jetson Orin Nano (SoC/heatsink)	OBC	FR4/ICs	80	7.00	Yes
STM32 Nucleo F446RE	OBC	FR4 PCB	50	0.30	—
SPOT Trace (upper deck)	Recovery	FR4 PCB	70	0.30	—
SPOT Trace (lower deck)	Recovery	FR4 PCB	70	0.30	—
Strato Tracker	Recovery	FR4 PCB	60	0.50	—
Flex 5 Powerbank	EPS	Li-ion cell	400	0.40	—
Wolfgang GA100 (frame camera)	Payload	Al-alloy	120	3.00	—
Prophesee EVK4 (IMX636)	Payload	Al-alloy	130	1.50	Yes

Table 4: Component operating temperature limits and data sources for power dissipation and limit values.

Component	$T_{op,min}$ [°C]	$T_{op,max}$ [°C]	Source
Battery (Li-ion / Li-Po)	−20	60	IEC 62133; Bernardi et al. 1985 <i>J. Electrochem. Soc.</i> 132(1):5–12

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Table 4 – continued

Component	$T_{op,min}$ [°C]	$T_{op,max}$ [°C]	Source
Jetson Orin Nano (board + SoC)	−25	90	NVIDIA DS-10653-001 Rev.1.0 (2023)
STM32 Nucleo F446RE	−40	85	ST UM1724 Rev.14; STM32F446RE datasheet
SPOT Trace (×2)	−40	60	SPOT Trace product specification
Strato Tracker	−40	85	Stratotracker product specification
Flex 5 Powerbank	−10	45	DC-DC buck 90–97%; IEC 62133
Wolfgang GA100 (frame camera)	−10	40	GoPro Hero 12/13 User Manual (2023) proxy
Prophesee EVK4 (IMX636)	−20	60	Prophesee Metavision EVK4 HD HW Guide Rev.C (2023)

4 Mission Profile

The analysed window is the powered ascent from launch to apogee and the first part of descent. The balloon ascends at 5 m/s to a burst altitude of 33 km ($t = 6600$ s), then descends under parachute at 5.5 m/s. The imaging altitude of 16 km is reached at $t \approx 3200$ s (53.5 min). In this operating concept the OBC (Jetson Orin Nano SoC, 7 W) and the event camera (EVK4, 1.5 W) are held unpowered during ascent and energised on reaching 16 km, remaining active for the one-hour imaging window to the end of the analysis ($t_{end} = 6800$ s); all other payload loads are active from launch. The resulting free-stream temperature and external convective coefficient are shown in Fig. 2: the ambient reaches the tropopause minimum near -56.5 °C, while h_{ext} falls monotonically with the thinning atmosphere.

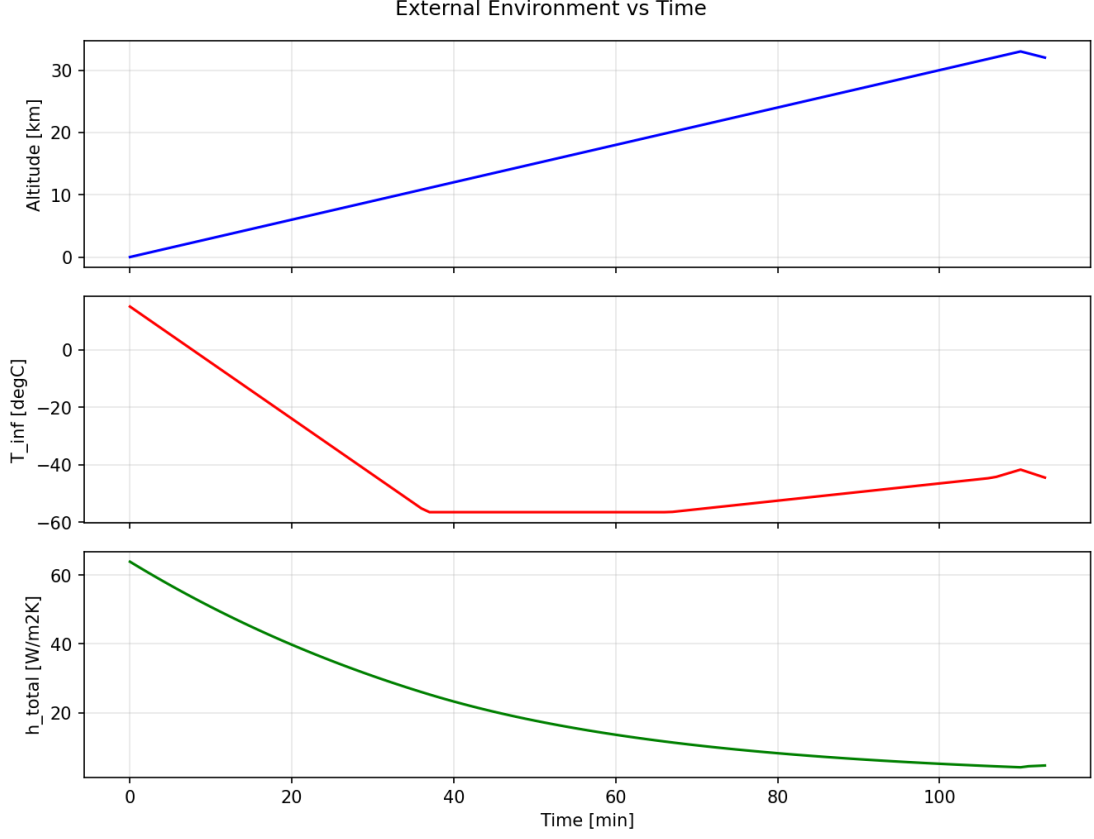


Figure 2: External environment over the mission window: altitude, free-stream temperature T_{∞} , and external convective coefficient h_{ext} from Block A. h_{ext} falls from $\approx 64 W/(m^2 K)$ at launch to below $5 W/(m^2 K)$ near apogee as the atmosphere thins.

5 Results

The internal temperature field is shown at five instants in Fig. 3. The cavity cools throughout the ascent while OBC loads are unpowered; on OBC energisation at 16 km the Jetson SoC (7 W) establishes a concentrated warm core in the bottom bay. Because the planar model resolves both lateral walls independently, the right-biased SoC heating pattern is captured directly in the contours without mirroring. Component temperature histories are given in Fig. 4. The full thermal margin summary is presented in Table 5.

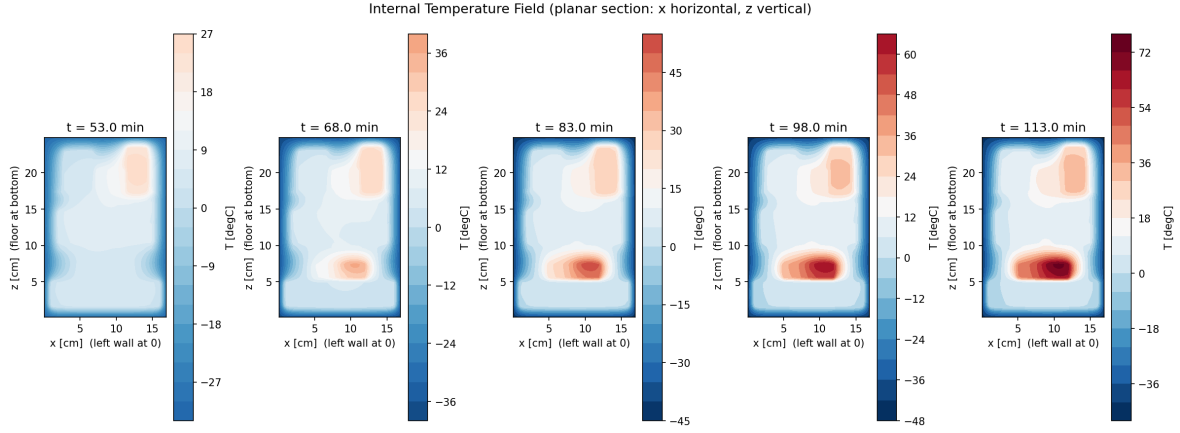


Figure 3: Internal temperature field (planar x - z section, full 170 mm cavity width) at five mission times. The cavity tracks the cooling wall during unpowered ascent; the warm bottom-row core from the Jetson SoC appears after OBC energisation at 16 km. Left-right asymmetry from the component layout is directly resolved.

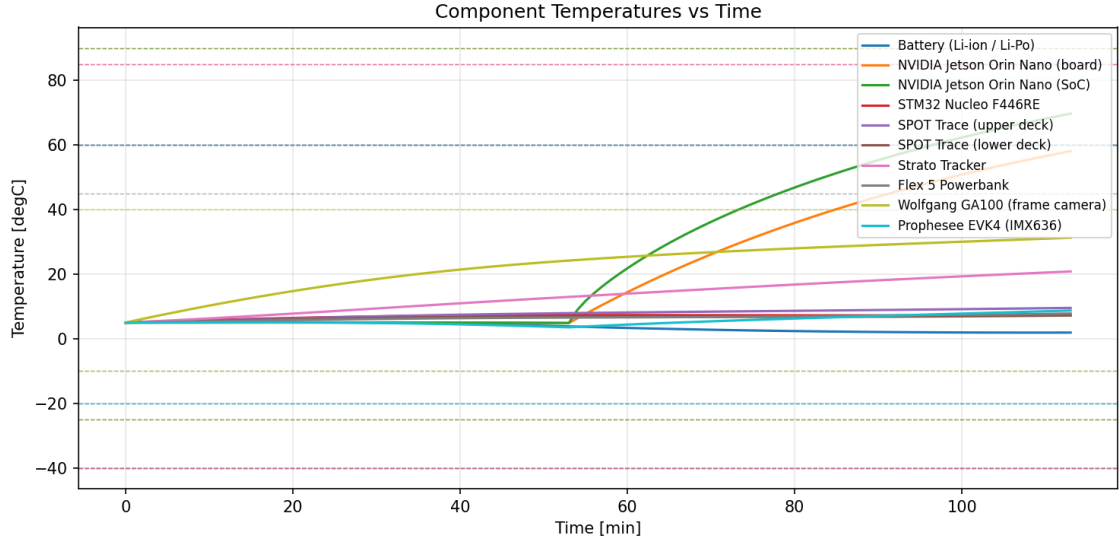


Figure 4: Component mean-temperature histories over the mission window with operating limit bands. OBC-gated components (Jetson SoC, EVK4) show a step rise at $t \approx 3200$ s on OBC energisation at 16 km.

Table 5: Thermal margin summary: simulation minima and maxima over the analysis window, operating limits, cold-side margin $\Delta T_{cold} = T_{min} - T_{op,min}$, and hot-side margin $\Delta T_{hot} = T_{op,max} - T_{max}$. All temperatures in $^{\circ}\text{C}$. All components within limits.

Component	T_{min}	T_{max}	$T_{op,min}$	$T_{op,max}$	ΔT_{cold}	ΔT_{hot}	Status
Battery (Li-ion / Li-Po)	1.9	5.1	-20	60	+21.9	+54.9	OK
Jetson Orin Nano (board)	4.8	58.1	-25	90	+29.8	+31.9	OK
Jetson Orin Nano (SoC)	4.9	69.8	-25	90	+29.9	+20.2	OK
STM32 Nucleo F446RE	5.0	7.4	-40	85	+45.0	+77.6	OK
SPOT Trace (upper deck)	5.0	9.5	-40	60	+45.0	+50.5	OK
SPOT Trace (lower deck)	5.0	7.2	-40	60	+45.0	+52.8	OK
Strato Tracker	5.0	20.8	-40	85	+45.0	+64.2	OK
Flex 5 Powerbank	5.0	7.8	-10	45	+15.0	+37.2	OK
Wolfgang GA100 (frame camera)	5.0	31.3	-10	40	+15.0	+8.7	OK
Prophesee EVK4 (IMX636)	3.5	8.7	-20	60	+23.5	+51.3	OK

The minimum cold-side margin is $+15.0^{\circ}\text{C}$, shared by the Flex 5 powerbank and the Wolfgang GA100 frame camera (both $T_{op,min} = -10^{\circ}\text{C}$); both are situated in the mid and upper enclosure rows and remain warm throughout the ascent. The tightest hot-side constraint is the GA100, which peaks at 31.3°C against a 40°C limit ($\Delta T_{hot} = +8.7^{\circ}\text{C}$), consistent with its steady 3 W dissipation throughout the mission. The Jetson SoC peaks at 69.8°C against a 90°C limit ($\Delta T_{hot} = +20.2^{\circ}\text{C}$) after OBC energisation. All remaining components have hot-side margins exceeding 20°C .

6 Conclusion

The coupled three-block thermal model, solved on a fully planar (x, z) Cartesian mesh without axial-symmetry assumptions and with a component set reconciled to the mechanical CAD bill of materials, shows that with the OBC powered at 16 km all ten payload components remain within their operating temperature limits throughout the ascent-to-apogee window from a $+5^{\circ}\text{C}$ initial temperature. The minimum cold-side margin is $+15.0^{\circ}\text{C}$ and all hot-side margins exceed 8°C , with the Wolfgang GA100 frame camera constituting the binding hot-side constraint. On this basis the payload thermal design is assessed **GO for flight**.